Astana IT University, School of Software Engineering
Pre-defense 1 report, version 2 (23 October 2026)

Development of a Machine Learning-Based System for Analyzing Patient Health Indicators

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Repository: github.com/MarioAman07/health-indicator-analysis-system · Tasks: YouTrack, project DIP

This version answers the supervisor's feedback of 23 October 2026. The main change is the focus. The report now describes the system as it would work in a Kazakhstani ICU: where the data come from, what happens every hour, how the nurse receives an alert and what happens next. It also shows how the diploma builds the same system on open data. Section 12 lists every comment and the change it led to. Full documents are in the repository; the links are given in each section.

1 Problem, aim and scope

Sepsis causes about 11 million deaths a year, one in five deaths worldwide [1], and treatment works best when it starts early [2]. In an ICU the early signs are in routine measurements recorded every hour, but they are spread over the monitor, the nurse's chart and the laboratory, and many values are missing. In Kazakhstan the escalation depends on a nurse noticing a change; both respondents said that early signs are missed at night, at shift handover and when one nurse has many patients, and one of them named the absence of a red-flag system. Systems that automate this step abroad are proprietary or built in-house.

Aim: design, build and evaluate an open decision-support system that estimates the risk of sepsis about six hours before onset from hourly ICU measurements and delivers each warning to the nurse with its reasons and a defined response.

Research questions:

- RQ1: does a gradient-boosted model (LightGBM) predict sepsis better than a partial SIRS rule and logistic regression? Hypothesis: higher AUPRC than logistic regression, 95% CI of the difference excluding 0.
- RQ2: how much does performance drop on a second hospital system? Expected: a drop for all models, reported for each.
- RQ3: do users interpret the warning correctly? Target: at least 2 of 3 respondents explain from the screen why the risk is high.

Scope: data validation, leakage-safe features, baselines and the main model, internal and external evaluation, alert rules, acknowledgement of alerts, audit, REST API, dashboard and tests. Out of scope: diagnosis or treatment advice, use on real patients, a working connection to a hospital system (designed, not built) and clinical validation.

Direction: after the meeting I chose the software-engineering direction, which matches my major. The model is evaluated honestly, but the depth of the work is in the design of the system, its fit with the clinical workflow and its verification.

2 Evidence of relevance

An anonymous survey (13 questions) was sent on 21 October to 8–10 people with clinical experience. Two answered and both answered five follow-up questions: R1, a ward nurse with more than five years of experience, and R2, a resident working in an ICU, a urology–nephrology department and the operating theatre. Their full answers are recorded as transcripts, in Russian and in translation, with the respondents' agreement: transcripts.md. Two respondents are below the target of 3–5, so the evidence is used to shape the design, not to describe practice in general.

What they said:

- A nurse notices a change during a routine measurement, checks the patient and informs the physician (R1, R2).

– ICU values are recorded every hour, ward values every 4–8 hours; they are read on the monitor or in the prescription sheet (R2).
– Changes are missed when early signs look minor, at handover, at night and with many patients; there is no red-flag system (R1, R2).
– A number alone is not enough: they want to see which measurements changed and when they were measured (R1, R2).
– Both chose: reasons for the risk, a warning about old data, a list sorted by risk. Both said the system must not replace the clinician.

They disagreed on three points, and the design takes both views. On data age, R1 would flag values older than 4–6 hours on a ward and R2 says that in an ICU values older than one hour are outdated, so the limit is a setting with a default of 4 hours. On how to show the risk, R2 prefers a percentage and R1 a colour with a short label, so both are shown. On repeated alerts, R1 wants no identical repeats and R2 wants an alert at every change, so repeats are suppressed but a new alert fires when the risk rises further.

What the data show. The PhysioNet/CinC Challenge 2019 data [3, 4] were analysed with a reproducible script (Table 2.1). Laboratory values are missing in 83–99.8% of hours, sepsis is rare at hour level (1.8%), the median onset is 43 hours after admission, so there is time to warn, and the two hospital systems record different laboratory values, which justifies an external test. Only 0.23% of gaps between key vital-sign measurements are longer than 4 hours, which supports the 4-hour default.

Table 2.1 – PhysioNet/CinC Challenge 2019 training data

	Set A	**Set B**	**Both**
ICU stays	20,336	20,000	40,336
Stays with sepsis	1,790 (8.80%)	1,142 (5.71%)	2,932 (7.27%)
Hours with SepsisLabel = 1	2.17%	1.42%	1.80%
Septic from the first hour	203	223	426

3 Literature review

Twelve peer-reviewed studies were read in full (search log in the literature matrix). Four findings shape the project.

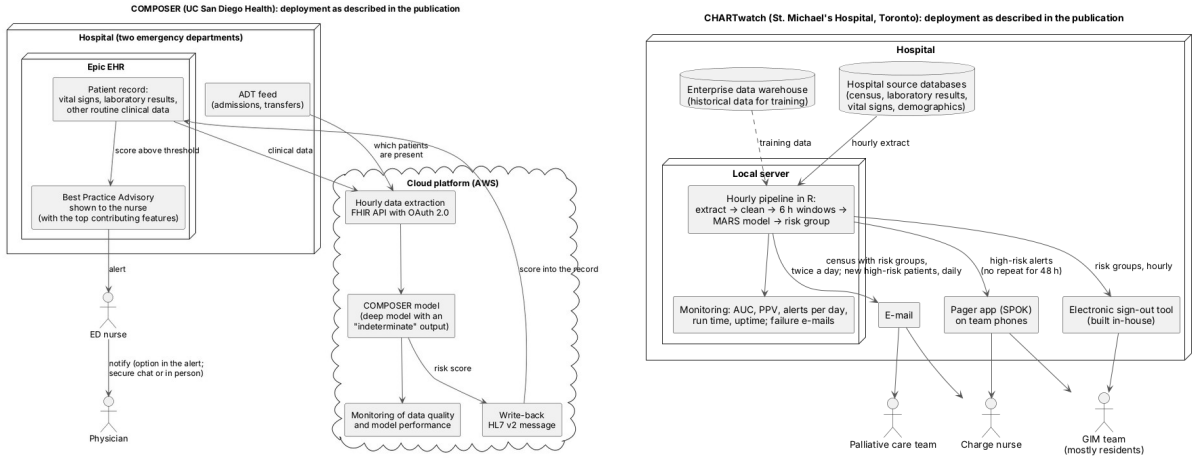
– **Machine learning does not always beat simple scores.** Gradient boosting outperformed NEWS while logistic regression did not [5]; a meta-analysis of 73 studies found high heterogeneity and risk of bias [6]; NEWS outperformed some proprietary AI scores [7]. The model is therefore compared with two baselines.
– **Performance drops at a new hospital.** Challenge rankings changed on an unseen hospital [3]; the Epic Sepsis Model had an external AUC of 0.63 and missed 67% of cases [8]; internal AUC 0.846 fell to 0.761 at new sites [9]. Set B is kept as an external test.
– **AUROC hides the alert burden.** Definitions and metrics differ too much to compare studies [10]; an "indeterminate" output cut false alarms by 78–86% [11]. The evaluation reports AUPRC, PPV and alerts per patient-day, and the system has an "insufficient data" state.
– **The benefit comes from the workflow.** The deployments with better outcomes combined the model with an alert and a defined human response [12, 13, 14], and performance falls from retrospective tests to live use [15]. The project designs the whole path from data to response and makes no clinical claims.

4 Existing systems and how they are built

Five deployed systems were compared on nine criteria (system comparison). None of them has a public ERD or an official architecture diagram, so the architectures in Figure 4.1 are reconstructed

from the publications; anything the sources do not describe is marked as such (system architectures).

- **COMPOSER** (UC San Diego): every hour a cloud platform pulls patient data from the Epic record through FHIR, computes the risk and writes it back as an HL7 message; the nurse sees an alert window with the main features and can notify the physician at once [12, 11].
- **CHARTwatch** (Toronto): an hourly R pipeline on a local server scores patients and sends high-risk alerts to the team's phones and risk lists to the charge nurse; repeats are blocked for 48 hours; it ran in silent mode before go-live and was available 99.6% of the time [16, 13].
- **TREWS:** a passive icon in the patient list opens a page with the reasons; a provider confirms or dismisses the alert, and timely confirmation was linked to lower mortality [17, 18].
- **Epic Sepsis Model and Deterioration Index:** closed models inside the vendor's record; only their inputs and thresholds are known from independent studies [8, 14].



a) COMPOSER *b) CHARTwatch*

Figure 4.1 – Reconstructed architectures of COMPOSER and CHARTwatch (TREWS and the Epic models are in the repository)

All of them take data from the hospital record on a schedule, compute the risk beside the record, return the result into tools the staff already use, make the staff's response part of the design, and monitor the system after go-live. My system follows the same pattern and differs in three ways: the data source is a replaceable adapter, the alert is acknowledged with a recorded outcome, and the code, model and evaluation are open.

5 How an ICU in Kazakhstan works today

Intensive care is regulated by the national Standard (Order No. 78, 2023) [19]: an ICU has at least six beds; the norm of 4.75 nurse posts per two beds means about one nurse on duty for every two patients; unstable patients get notes every 3 hours. The respondents describe hourly recording of heart rate, blood pressure, saturation, respiratory rate and temperature. Hospitals keep records in a medical information system (MIS) [20]; one of them, Damumed, covered more than 80% of the market in 2023 [21]. Since July 2026 the requirements for medical digital systems allow HL7 integration with laboratory systems and monitoring devices and require hosting in Kazakhstan, role-based access and logging of user actions [22].

When a nurse notices a change, the nurse checks the patient and informs the physician, who follows the national protocol "Sepsis" [23]: lactate and blood cultures within 1.5 hours, fluids within the first hour, antibiotics without delay, source control within 6 hours (Figure 5.1). The protocol starts once sepsis is suspected. The weak point is before that moment, and I found no early-warning tool in the Standard or the protocol.

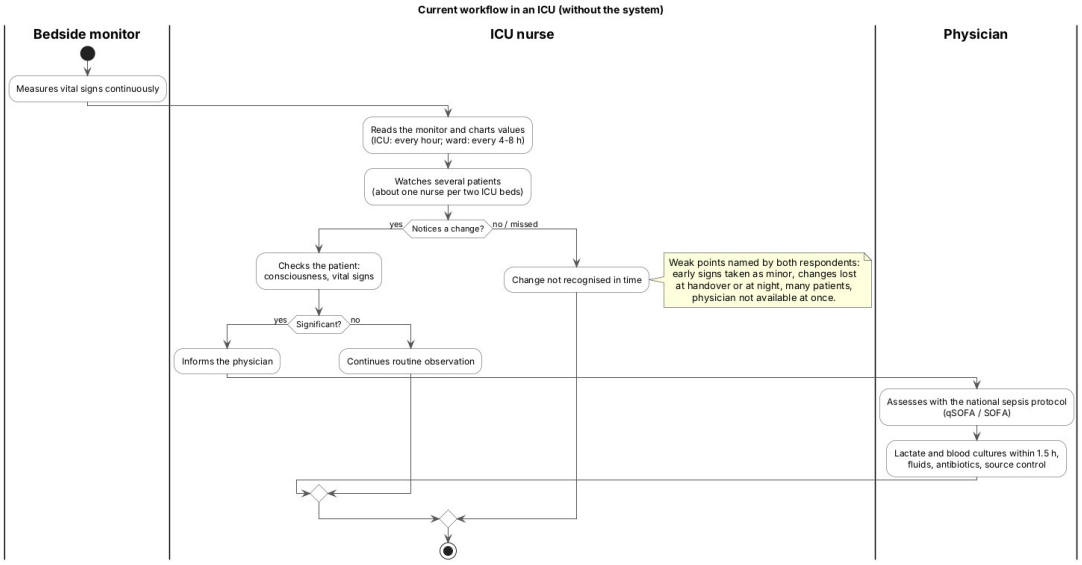


Figure 5.1 – Current workflow, without the system

The place of the system is therefore narrow: look at every patient every hour in the same way, notice a combination of changes that is easy to miss at night or at handover, and tell the nurse early, with the reasons. Everything after that stays with the nurse, the physician and the protocol.

6 How the system works

Full description: target-system.md.

6.1 Where the data come from

In a hospital the system reads from the MIS the vital signs charted by the nurse every hour, the laboratory results and the admission data, through HL7 v2 messages (ADT, ORU) or the MIS interface. Charted values are used rather than raw monitor signals, because the nurse has already checked them. The system measures nothing itself; it derives indicators from what was recorded: the last value, the time since it was measured and its change over 6 and 12 hours. The nurse sees the same values next to the risk.

The PhysioNet files have the same shape, one row per patient-hour. In the diploma a replay adapter plays them hour by hour and at hour t releases only rows with ICULOS $\leq t$, so the system cannot see the future. A hospital would replace only this adapter.

6.2 What happens every hour

1) Read the new values from the data source.
2) Validate them; a wrong value is logged and skipped.
3) If no key vital sign was recorded in the last 4 hours, show "insufficient data" and ask for a measurement; no risk, no alert.
4) Build features from past hours only and compute the risk with the five main reasons.
5) Apply the alert rules: alert at the threshold; after any alert, no repeat for 6 hours unless the risk rises by 10 points.
6) Store the prediction, the alert and an audit record; the dashboard refreshes every minute.

If the data source fails, nothing new is computed and the dashboard shows the time of the last successful cycle; after 4 hours patients pass into "insufficient data", so a broken feed is visible instead of old risks staying on the screen. Figure 6.1 shows the cycle and the response to one alert.

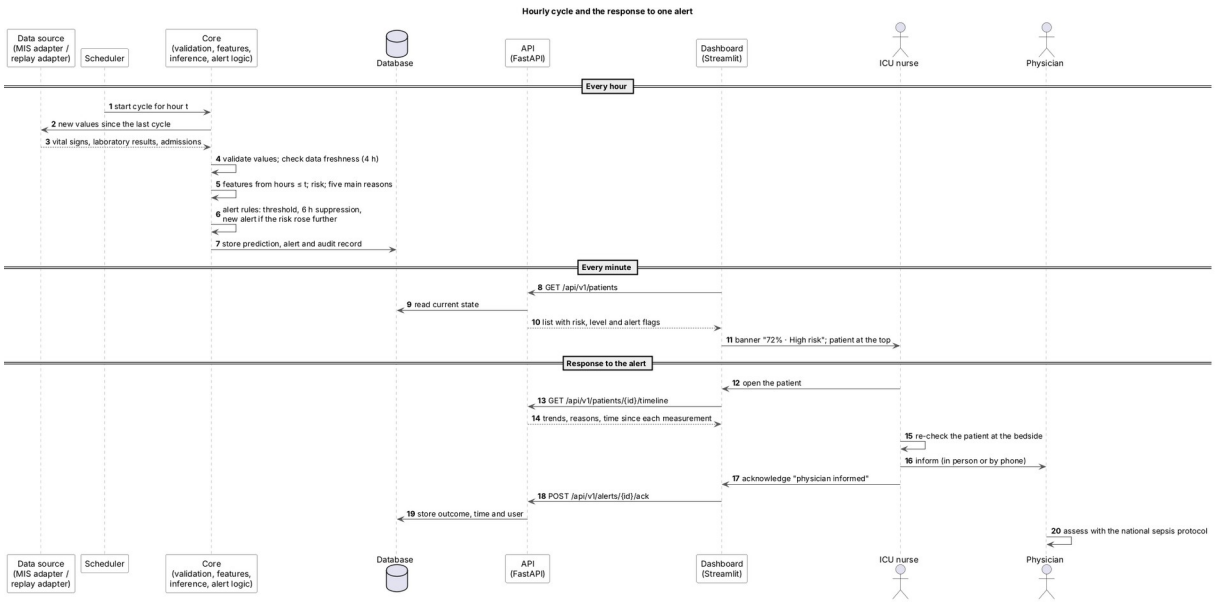


Figure 6.1 – Hourly cycle and the response to one alert

6.3 What the nurse sees and does

The patient list at the nurse station is sorted by urgency: unacknowledged alerts, acknowledged alerts, the others by risk, insufficient data last. Each row shows the risk as "72% · High risk" in red, yellow or green. A new alert shows a banner until it is acknowledged. The patient view shows the risk and five vital signs on one time axis, the time since each measurement, the five reasons and one fixed prompt: "Re-check the measurements and inform the responsible physician". The system gives no treatment advice.

After the alert the nurse re-checks the patient at the bedside and acknowledges the alert with one of three outcomes: physician informed, change not confirmed, or already being treated (Figure 6.2). The physician then acts under the national protocol. An alert not acknowledged within 30 minutes stays at the top marked "not acknowledged". The share of "change not confirmed" is a direct local measure of false alerts.

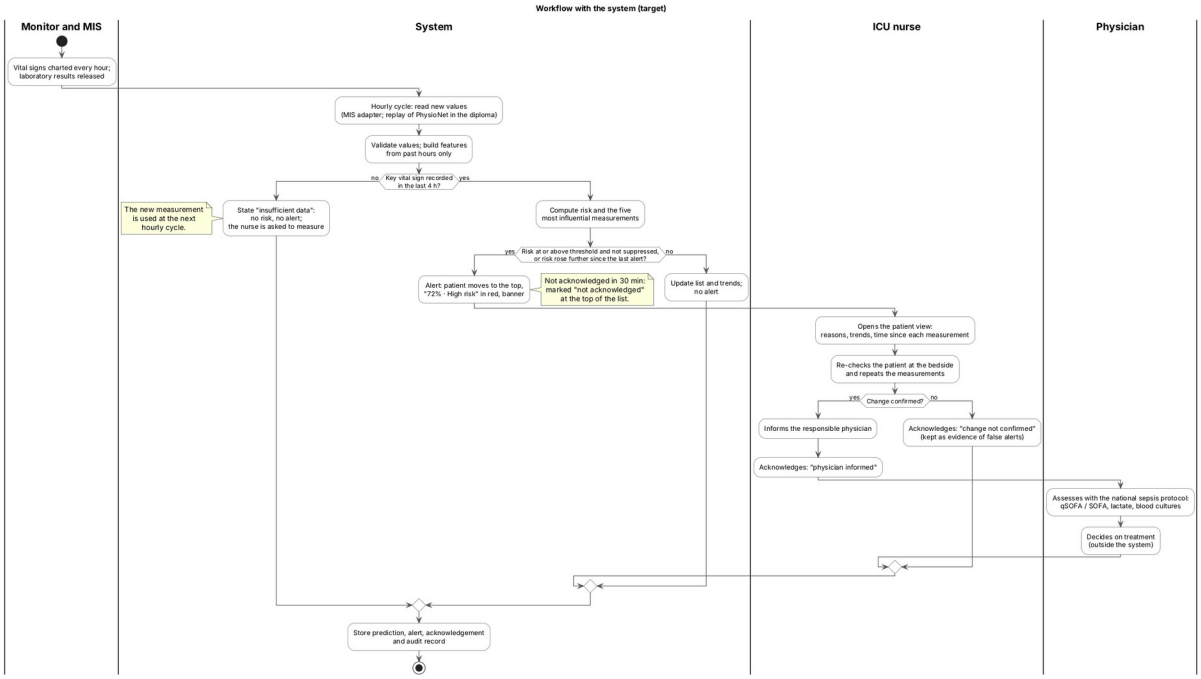


Figure 6.2 – Workflow with the system

6.4 Deployment and introduction

In a hospital everything runs on an on-premises server in Kazakhstan: the MIS adapter, a scheduler, the FastAPI backend with the model, PostgreSQL and the dashboard over HTTPS, with role-based login (nurse, physician, administrator) and an audit trail. In the diploma the same components run on one laptop with the replay adapter and SQLite (Figure 6.3). A hospital would introduce it in three steps, as CHARTwatch did [16]: silent mode for several weeks, a pilot on one unit with staff training, then review of alert frequency, acknowledgement times and false alerts.

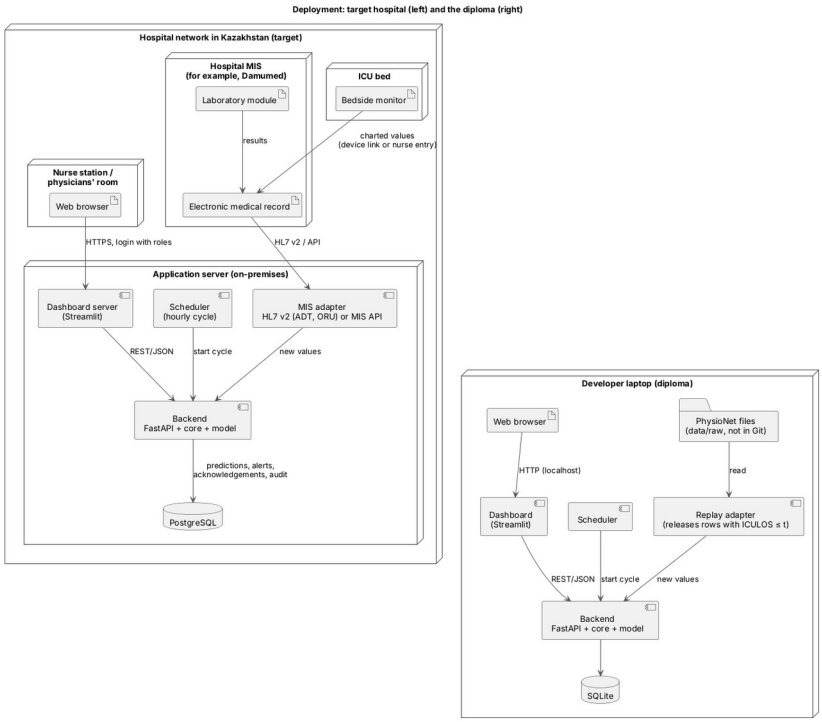
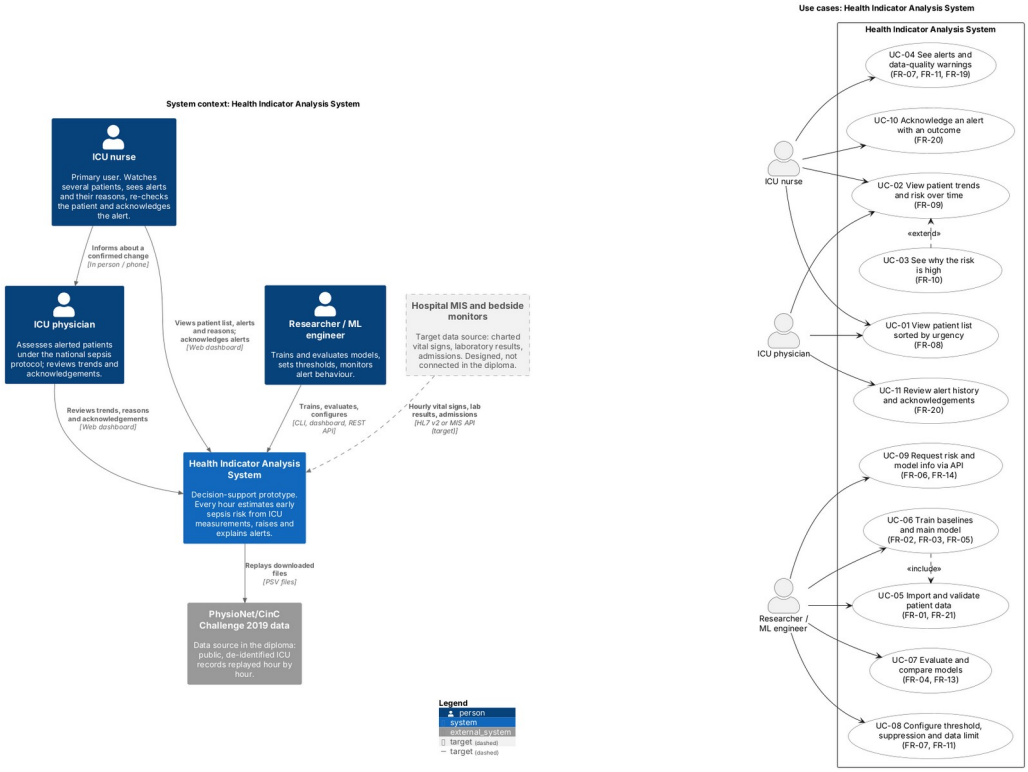


Figure 6.3 – Deployment in a hospital and in the diploma

Built in the diploma: the data-source interface with replay, the hourly cycle, the model, all alert rules, acknowledgement, audit, API and dashboard. Designed only, because they need a real hospital: the MIS adapter, phone or MIS notifications, hospital accounts, silent mode and pilot.

7 Architecture, data model and technology

The system is a modular monolith in four layers: dashboard, REST API, core package (data source, validation, features, model, alert logic, explanations) and storage (Figures 7.1–7.3). The dashboard only calls the API, so all behaviour can be tested through the API, and training and live prediction use the same feature code. Details: architecture.md.



a) Context *b) Use cases*

Figure 7.1 – System context and use cases

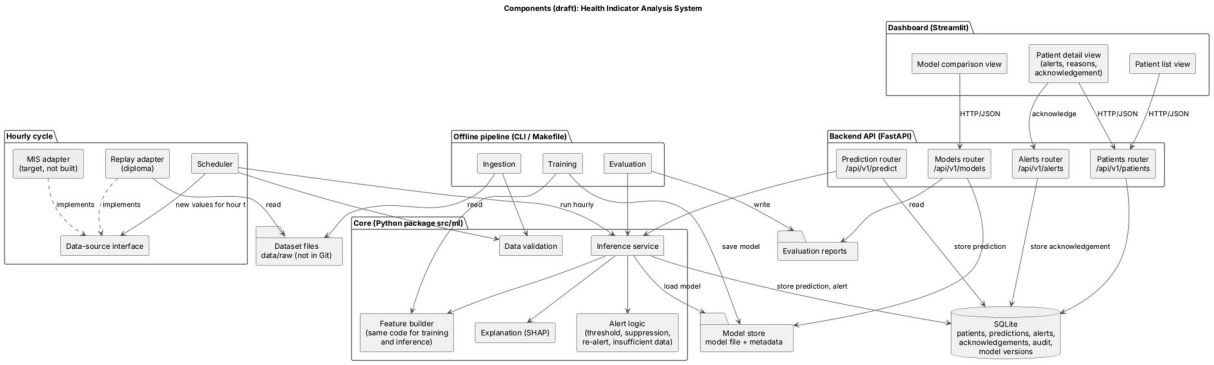


Figure 7.2 – Components

The data model has eight entities (Figure 7.3): a Patient has hourly Measurements; each cycle writes a Prediction made by a ModelVersion; a Prediction may raise an Alert; the nurse closes it with an Acknowledgement; every view, acknowledgement and setting change leaves an AuditRecord made by a User.

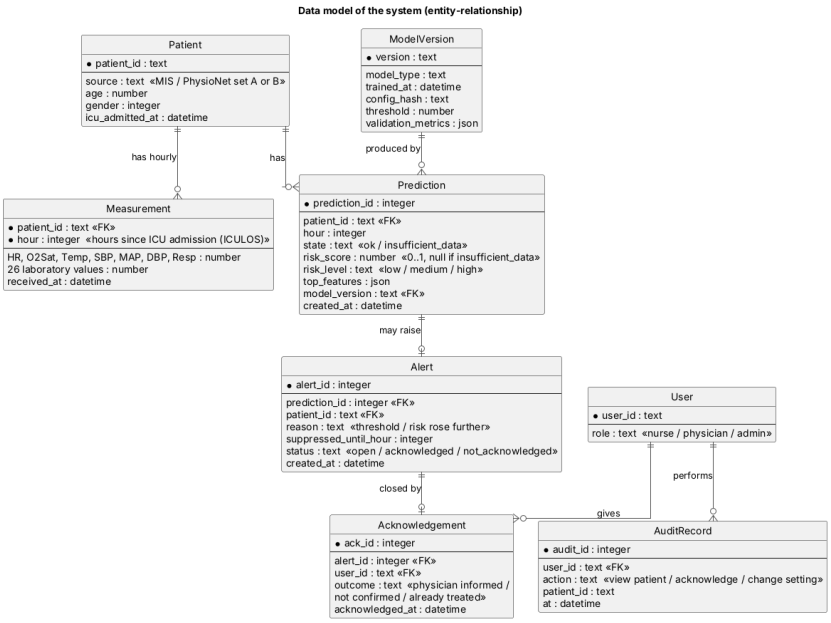


Figure 7.3 – Data model (ERD)

Technology choices. FastAPI instead of Flask or Django: typed request validation and an automatic OpenAPI description, with the model in the same Python process. REST/JSON instead of gRPC or a message queue: one client and one backend need simple request–response calls. SQLite in the diploma and PostgreSQL in a hospital, behind SQLAlchemy, so the change does not touch the code. Streamlit instead of React: the dashboard stays in Python and contains no logic. An hourly batch instead of streaming, because the ICU records values every hour. Full comparison: technology-comparison.md.

Machine-learning method (ml-methodology.md). Data: 40,336 ICU stays from two hospital systems. Label: `SepsisLabel`, set 6 hours before Sepsis-3 onset. Split by patient: set A 70/15/15 for training, validation and internal test; set B only as the external test. Features use hours ≤ *t* only (checked by a unit test): last values, missingness and time since measurement, 6 h and 12 h trends of six vital signs, age and ICU hour. Models: partial SIRS rule, logistic regression, LightGBM with SHAP explanations. Metrics: AUROC, AUPRC, PPV at 80% sensitivity, calibration, alerts per patient-day and lead time, with bootstrap confidence intervals.

8 Requirements

The specification has 19 functional requirements to build (10 Must, 7 Should, 2 Could), 2 excluded ones and 10 non-functional requirements, each with a pass/fail acceptance criterion and a source: literature, interviews, data or supervisor (06-requirements.md). After the meeting I added FR-19 – FR-21 and NFR-10 and changed FR-07, FR-08, FR-09 and FR-11 (Table 8.1).

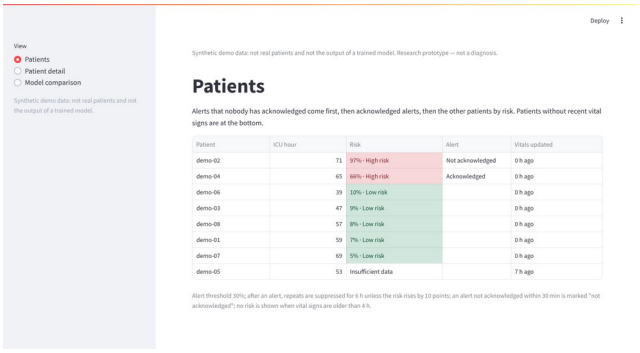
Table 8.1 – Requirements added or changed after 23 October

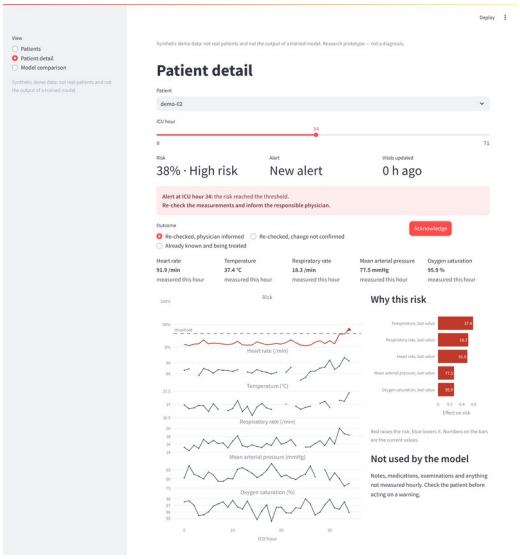
ID	Requirement	Priority
FR-07	Alert at the threshold; no repeat for 6 h after an alert unless the risk rises by 0.10	Must
FR-08	Patient list by urgency; risk as a percentage with a coloured label	Must
FR-09	Trends with the time since each measurement	Must
FR-11	"Insufficient data" when key vital signs are older than 4 h (configurable)	Should
FR-19	One fixed prompt with every alert	Should
FR-20	Acknowledgement with three outcomes; "not acknowledged" after 30 min	Should
FR-21	Data-source interface, replay adapter and hourly cycle	Must
NFR-10	Audit records of views, acknowledgements and setting changes	Should

The other Must requirements cover file validation (FR-01), no future data in features (FR-02), one-command training (FR-03), evaluation on both test sets (FR-04), the partial SIRS score (FR-05), the prediction API (FR-06), reproducibility, data protection, portability and the disclaimer "Research prototype — not a diagnosis". The 30-minute limit and the 0.10 margin are first values, to be checked with ICU staff.

9 Prototype

A clickable Streamlit prototype runs on synthetic data and applies the same alert rules as the planned system (Figure 9.1). It shows the list by urgency, the alert banner with the fixed prompt, the acknowledgement, the time since each measurement and the "insufficient data" state. The API contract has eight endpoints, including `POST /api/v1/alerts/{id}/ack` for acknowledgements (api-examples.md). Known gap: the patient view still needs scrolling at 1366×768 (NFR-07).





a) Patient list *b) Patient view with an alert*

Figure 9.1 – Prototype screens (synthetic data)

10 Verification and evaluation plan

Every Must and Should requirement has a test: unit tests (for example, no future data in features and the alert rules with the exact values of FR-07), integration tests on a fixed sample of 50 patients, API tests, acceptance tests and inspections (09-verification.md). Twelve tests run today: six for the data-analysis script and six for the alert and data rules of the prototype.

The model is evaluated on the internal test (15% of set A) and the external test (set B) against the two baselines. Success: LightGBM has a higher AUPRC than logistic regression with a 95% CI of the difference excluding 0; the drop on set B is reported for every model. The workflow is checked with respondents: can they explain a high risk from the screen, is 30 minutes realistic for acknowledgement, and is the re-alert margin right.

11 Project management and risks

Every task is a YouTrack issue with a branch `DIP-<n>-<name>`, commits starting with `DIP-<n>:` and a pull request into the protected `main`. Before 19 October some pull requests were merged without review; this is recorded and the rules apply from 21 October.

Table 11.1 – Milestones (dates after pre-defense 1 are proposed)

Milestone	Date	Done when
Pre-defense 1	24 Oct 2026	All 11 components merged, tag pushed, form submitted
M1 Data pipeline and baselines	20 Nov 2026	Validation, leakage test, replay adapter, hourly cycle, baselines
M2 Main model and evaluation	18 Dec 2026	Internal and external results with CIs, model card
M3 API and dashboard	26 Feb 2027	Acceptance tests incl. acknowledgement and audit; user check
M4 Testing and documentation	26 Mar 2027	All Must tests pass; clean-setup demonstration

Main risks: too few respondents (survey continues; the actual number is reported); temporal leakage (patient split, past-only features, unit test, replay adapter); LightGBM not beating logistic regression or dropping on set B (reported as results); the workflow not matching real ICU practice (checked with ICU staff, silent mode first); no access to a hospital MIS (the adapter is designed, not claimed as built). Full register: 14 risks in the repository.

12 Supervisor feedback and what changed

All feedback is recorded in the feedback log. The written review of 19 October (README, problem and users, own vision, ML justification, critical comparison, missing components) was addressed before the meeting. Table 12.1 shows the comments of 23 October.

Table 12.1 – Feedback of 23 October and the changes

Comment	What changed	Where
Focus on the system as it should be; show how the whole process will run	Target system described separately from the diploma implementation and connected through the data-source interface	Sections 6–7; target-system.md (DIP-27)
How the ICU works in Kazakhstan; what happens after a notification	ICU organisation, records and the sepsis protocol; step-by-step response with acknowledgement (FR-20)	Sections 5, 6.3
How alerts and data are shown; how indicators are measured; where data come from	Data sources, hourly cycle, screens; five changes from the interviews adopted; prototype updated	Sections 6, 8, 9 (DIP-27, DIP-30)
Show how existing systems are built (ERD, architecture)	No public ERDs exist; architectures reconstructed from publications	Section 4; system-architectures.md (DIP-28)
Record respondents' answers as transcripts	Full answers in Russian and English, published with consent	Section 2; transcripts.md (DIP-26)
Report: more argued, fewer tables, simpler and more academic	Report rewritten in this form	This report (DIP-29)
Choose: well-developed system (SE) or deep ML	SE direction chosen	Section 1

Still open: my own summary of the reasoning behind the technical decisions (FB-08), more respondents and the user check of the prototype.

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